

POWERFACTORY

PowerFactory 2024

Technical Reference

Power Measurement

StaPqmea

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

The *Power Measurement Device* (*StaPqmea*) can be used to measure the power flow at a cubicle of any element, which is connected to a terminal/busbar. The measured active and reactive power can then be fed for example into a DSL model of a controller.

The power direction (sign) can be defined for the measurement device individually according to the parameter *Orientation* (*i_orient*):

- Load oriented
- Generator oriented
- Orientation of connected element

If the third option is selected, the direction of the power flow depends on the element where the power is being measured. For loads, motors or passive elements (lines, transformers, etc.) the power flow is defined in load orientation. This means the power which flows out of the terminal into the element is positive. For sources, e.g. generators, external network, current and voltage sources, etc. the power flow is in generator orientation and thus out of the element in the direction of the connected bus is defined as positive.

The power output is normalised depending on the selected *Power Rating* (i_mode parameter) to:

- *Equivalent to MVA value* where the p.u. output value corresponds to the values in MVA and Mvar,
- *Power rating of connected element* where the p.u. output is based on the power rating of the connected element, and
- *User-defined power rating* where the p.u. output is calculated using the $Snom$ input parameter as base value.

For the EMT calculation, an option is also available to measure RMS quantities. This option can be selected on the EMT page.

The measurement device is mostly used in combination with controller models. If an element and its controller models are using the A-stable integration algorithm, the connected measurement device should also use this algorithm, thus the flag *iAstabint* on the RMS-simulation page should be selected.

The equations and parameters of the *StaPqmea* element are documented in the following sections.

2 Load Flow

The output signals available for the Load Flow calculation are identical to the output signals presented in the RMS-Simulation. Please refer to Section 3 for more information.

3 RMS-Simulation

The equations for balanced and unbalanced RMS simulation are described in the following subsections.

3.1 Balanced (symmetrical components)

From the balanced RMS-Simulation, the real and imaginary part of the positive-sequence voltage are available.

- For AC systems, the power output signals are calculated as follows:

$$p = \mathrm{Re}\{3 \cdot \underline{u}_1 \cdot \underline{i}_1^*\} \quad (1)$$

$$q = \mathrm{Im}\{3 \cdot \underline{u}_1 \cdot \underline{i}_1^*\} \quad (2)$$

  where * is the complex conjugate operator.

- For DC systems, only the real parts of the voltage and current are available:

$$p = u_1 \cdot i_1 \quad (3)$$

$$q = 0 \quad (4)$$

The output signals of the balanced RMS model are presented in Table 3.1.

Table 3.1: Outputs, Balanced (symmetrical components) RMS-Simulation

Name	Unit	Description
p	p.u.	Positive sequence active power
q	p.u.	Positive sequence reactive power

3.2 Unbalanced (symmetrical components)

The three-phase unbalanced RMS simulation delivers the complex voltages and currents for phase A, B and C to the measurement device ($\underline{u}_a$, $\underline{u}_b$, $\underline{u}_c$, $\underline{i}_a$, $\underline{i}_b$ and $\underline{i}_c$) . The output power signals for three-phase unbalanced RMS simulation are calculated depending on the phase technology of the element. The unbalanced measurement depends on the *No. of Phases* option (parameter $nphase$).

In the case of a three-phase system, the output signals are the active and reactive power which are calculated using the positive sequence of the voltage and current. Similar calculation is done for the BI and two phase system.

- 3-phase system (using symmetrical components transformation):

The sequence voltages and currents are calculated as:

$$\underline{u}_0 = \frac{1}{3} \cdot [\underline{u}_a + \underline{u}_b + \underline{u}_c] \tag{5}$$

$$\underline{u}_1 = \frac{1}{3} \cdot \left[\underline{u}_a + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) \cdot \underline{u}_b + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) \cdot \underline{u}_c\right] \tag{6}$$

$$\underline{u}_2 = \frac{1}{3} \cdot \left[\underline{u}_a + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) \cdot \underline{u}_b + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) \cdot \underline{u}_c\right] \tag{7}$$

$$\underline{i}_0 = \frac{1}{3} \cdot [\underline{i}_a + \underline{i}_b + \underline{i}_c] \tag{8}$$

$$\underline{i}_1 = \frac{1}{3} \cdot \left[\underline{i}_a + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) \cdot \underline{i}_b + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) \cdot \underline{i}_c\right] \tag{9}$$

$$\underline{i}_2 = \frac{1}{3} \cdot \left[\underline{i}_a + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) \cdot \underline{i}_b + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) \cdot \underline{i}_c\right] \tag{10}$$

The output powers are calculated as:

$$p0 = \mathrm{Re}\{\sqrt{2} \cdot 3 \cdot \underline{u}_0 \cdot \underline{i}_0^*\} \tag{11}$$

$$q0 = \mathrm{Im}\{\sqrt{2} \cdot 3 \cdot \underline{u}_0 \cdot \underline{i}_0^*\} \tag{12}$$

$$p = \mathrm{Re}\{\sqrt{2} \cdot 3 \cdot \underline{u}_1 \cdot \underline{i}_1^*\} \tag{13}$$

$$q = \mathrm{Im}\{\sqrt{2} \cdot 3 \cdot \underline{u}_1 \cdot \underline{i}_1^*\} \tag{14}$$

$$p2 = \mathrm{Re}\{\sqrt{2} \cdot 3 \cdot \underline{u}_2 \cdot \underline{i}_2^*\} \tag{15}$$

$$q2 = \mathrm{Im}\{\sqrt{2} \cdot 3 \cdot \underline{u}_2 \cdot \underline{i}_2^*\} \tag{16}$$

- BI-phase system ($180°$)

  The sequence voltages and currents are calculated as:

$$\underline{u}_0 = \frac{1}{2} \cdot (\underline{u}_a + \underline{u}_b) \tag{17}$$

$$\underline{u}_1 = \frac{1}{2} \cdot (\underline{u}_a - \underline{u}_b) \tag{18}$$

$$\underline{i}_0 = \frac{1}{2} \cdot (\underline{i}_a + \underline{i}_b) \tag{19}$$

$$\underline{i}_1 = \frac{1}{2} \cdot (\underline{i}_a - \underline{i}_b) \tag{20}$$

  The output powers are calculated as:

$$p = \mathrm{Re}\{2 \cdot \underline{u}_1 \cdot \underline{i}_1^*\} \tag{21}$$

$$q = \mathrm{Im}\{2 \cdot \underline{u}_1 \cdot \underline{i}_1^*\} \tag{22}$$

$$p0 = \mathrm{Re}\{2 \cdot \underline{u}_2 \cdot \underline{i}_2^*\} \tag{23}$$

$$q0 = \mathrm{Im}\{2 \cdot \underline{u}_2 \cdot \underline{i}_2^*\} \tag{24}$$

- 2-phase system ($120°$)

The sequence voltages and currents are calculated as:

$$\underline{u}_0 = \frac{1}{\sqrt{3}} \cdot (\underline{u}_a + \underline{u}_b) \tag{25}$$

$$\underline{u}_1 = \frac{1}{\sqrt{3}} \cdot (\underline{u}_a - \underline{u}_b) \tag{26}$$

$$\underline{i}_0 = \frac{1}{\sqrt{3}} \cdot (\underline{i}_a + \underline{i}_b) \tag{27}$$

$$\underline{i}_1 = \frac{1}{\sqrt{3}} \cdot (\underline{i}_a - \underline{i}_b) \tag{28}$$

$$\tag{29}$$

The output powers are calculated as:

$$p = \mathrm{Re}\left\{\frac{3}{2} \cdot \underline{u}_1 \cdot \underline{i}_1^*\right\} \tag{30}$$

$$q = \mathrm{Im}\left\{\frac{3}{2} \cdot \underline{u}_1 \cdot \underline{i}_1^*\right\} \tag{31}$$

$$p0 = \mathrm{Re}\left\{\frac{3}{2} \cdot \underline{u}_2 \cdot \underline{i}_2^*\right\} \tag{32}$$

$$q0 = \mathrm{Im}\left\{\frac{3}{2} \cdot \underline{u}_2 \cdot \underline{i}_2^*\right\} \tag{33}$$

- Single-phase (system):

  The output powers are calculated as:

$$p = \mathrm{Re}\left\{\underline{u}_a \cdot \underline{i}_a^*\right\} \tag{34}$$

$$q = \mathrm{Im}\left\{\underline{u}_a \cdot \underline{i}_a^*\right\} \tag{35}$$

  In case of a multi phase system where only the power of one phase should be measured the option *No. of Phases* has to be set to 1. Depending on the selected *Measurement Point*, the power can be measured at phase *a, b, c, a-n, b-n or c-n* by selecting the appropriate *Measured Phase*.

  The output signals are calculated as above.

- DC system:

$$p = u_a \cdot i_a \tag{36}$$

$$q = 0 \tag{37}$$

The output signals of the unbalanced RMS model are presented in Table 3.2.

Table 3.2: Outputs, Unbalanced (symmetrical components) RMS-Simulation

Name	Unit	Description
p	p.u.	Positive sequence active power
q	p.u.	Positive sequence reactive power
$p2$	p.u.	Negative sequence active power
$q2$	p.u.	Negative sequence reactive power
$p0$	p.u.	Zero sequence active power
$q0$	p.u.	Zero sequence reactive power

4 EMT-Simulation

Same as in the RMS simulation, two types of measurement modes are available depending on the *No. of Phases* parameter.

In the EMT simulation, in addition the RMS quantities can be also calculated.

4.1 Alpha, beta and gamma components

The three-phase unbalanced EMT simulation provides the phase voltages and currents for phase A, B and C to the measurement device (u_a, u_b, u_c, i_a, i_b and i_c).

The output power signals for EMT simulation are calculated as follows:

- 3-phase system:

  The phase voltages and currents are first transformed using the $\alpha\beta\gamma$ transformation.

$$u_\alpha = \frac{1}{3} \cdot (2 \cdot u_a - u_b - u_c) \tag{38}$$

$$u_\beta = \frac{1}{3} \cdot (\sqrt{3} \cdot u_b - \sqrt{3} \cdot u_c) \tag{39}$$

$$u_\gamma = \frac{1}{3} \cdot (u_a + u_b + u_c) \tag{40}$$

$$i_\alpha = \frac{1}{3} \cdot (2 \cdot i_a - i_b - i_c) \tag{41}$$

$$i_\beta = \frac{1}{3} \cdot (\sqrt{3} \cdot i_b - \sqrt{3} \cdot i_c) \tag{42}$$

$$i_\gamma = \frac{1}{3} \cdot (i_a + i_b + i_c) \tag{43}$$

  The output signals are calculated as follows:

$$p = \mathrm{Re}\left\{\frac{3}{2} \cdot (u_\alpha + j\, u_\beta) \cdot (i_\alpha - j\, i_\beta)\right\} \tag{44}$$

$$q = \mathrm{Im}\left\{\frac{3}{2} \cdot (u_\alpha + j\, u_\beta) \cdot (i_\alpha - j\, i_\beta)\right\} \tag{45}$$

$$p0 = \frac{3}{2} \cdot u_\gamma \cdot i_\gamma \tag{46}$$

  Note: the reactive power calculation is accurate under the assumption of balanced voltages and currents. Carrying out the calculations, one finds that:

$$q = \frac{3}{2} \cdot (u_\beta \cdot i_\alpha - u_\alpha \cdot i_\beta) \tag{47}$$

$$q = \frac{1}{\sqrt{3}} \cdot \{(u_b - u_c) \cdot i_a + (u_c - u_a) \cdot i_b + (u_a - u_b) \cdot i_c\} \tag{48}$$

- BI-phase system ($180°$):

The phase voltages and currents are first transformed using the $\alpha\beta\gamma$ transformation.

$$u_\alpha = \frac{1}{2} \cdot (u_a - u_b) \tag{49}$$

$$u_\gamma = \frac{1}{2} \cdot (u_a + u_b) \tag{50}$$

$$i_\alpha = \frac{1}{2} \cdot (i_a - i_b) \tag{51}$$

$$i_\gamma = \frac{1}{2} \cdot (i_a + i_b) \tag{52}$$

The output signals are calculated as follows:

$$p = 2 \cdot u_\alpha \cdot i_\alpha \tag{53}$$

$$q = 0 \tag{54}$$

$$p0 = 2 \cdot u_\gamma \cdot i_\gamma \tag{55}$$

- 2-phase system ($120°$):

  The phase voltages and currents are first transformed using the $\alpha\beta\gamma$ transformation.

$$u_\alpha = \frac{1}{\sqrt{3}} \cdot (u_a - u_b) \tag{56}$$

$$u_\gamma = \frac{1}{\sqrt{3}} \cdot (u_a + u_b) \tag{57}$$

$$i_\alpha = \frac{1}{\sqrt{3}} \cdot (i_a - i_b) \tag{58}$$

$$i_\gamma = \frac{1}{\sqrt{3}} \cdot (i_a + i_b) \tag{59}$$

  The output signals are calculated as follows:

$$p = \frac{3}{2} \cdot u_\alpha \cdot i_\alpha \tag{60}$$

$$q = 0 \tag{61}$$

$$p0 = \frac{3}{2} \cdot u_\gamma \cdot i_\gamma \tag{62}$$

- Single-phase (system):

  The output signals are calculated as follows:

$$p = u_a \cdot i_a \tag{63}$$

$$q = 0 \tag{64}$$

$$p0 = 0 \tag{65}$$

  For the unbalanced EMT simulation, depending on the selected *Measurement Point*, the power can be measured at phase *a, b, c, a-n, b-n and c-n*.

  In this case the output signals are calculated as above.

- DC system:

  The output signals are calculated as follows:

$$p = u_a \cdot i_a \tag{66}$$

$$q = 0 \tag{67}$$

$$p = 0 \tag{68}$$

The output signals used by the EMT model using the Alpha, beta and gamma components are presented in Table 4.1.

Table 4.1: Outputs, Alpha, beta and gamma components EMT-Simulation

Name	Unit	Description
p	p.u.	Active power
q	p.u.	Reactive power
$p0$	p.u.	Active Power, gamma-component

4.2 RMS phasor and symmetrical components (phase quantities)

The measurement device measuring RMS phasors is a clocked device which calculates a Discrete Fourier Transform (DFT) for a single frequency. There are four different options for estimating the frequency for which the DFT is calculated:

- *Use nominal frequency*

  The DFT is executed for nominal frequency.

- *Phase angle derivative*

  The frequency supplied to the DFT (fe), is estimated based on the phase angle derivative.

  The calculation of the frequency fe is similar to the calculation in the RMS simulation, with the difference that instead the positive-sequence, the dq voltage values are used. The dq quantities of the voltage are obtained as:

$$ud + j\ uq = (ur + j\ ui) \cdot (\cos(phiref) - j\ (\sin(phiref))) \tag{69}$$

  The following differential equations are being used:

$$\frac{d\ \cos\phi}{dt} = \left(\frac{ud}{\sqrt{ud^2 + uq^2}} - \cos\phi\right) \Big/ T_{fe} \tag{70}$$

$$\frac{d\ \sin\phi}{dt} = \left(\frac{uq}{\sqrt{ud^2 + uq^2}} - \sin\phi\right) \Big/ T_{fe} \tag{71}$$

  where T_fe is fixed to $3/Fnom$.

  The frequency change (in p.u.) is equal to:

$$dfe = \sqrt{\frac{\left(\frac{d\ \cos\phi}{dt}\right)^2 + \left(\frac{d\ \sin\phi}{dt}\right)^2}{(\cos\phi)^2 + (\sin\phi)^2}} \Big/ (2 \cdot \pi \cdot Fnom) \tag{72}$$

  The frequency is then calculated as $fe = 1 + dfe$ or as $fe = 1 - dfe$ depending on the angle change.

  Please note that this option works best with balanced, sinusoidal signals and is available only for three-phase systems.

- *Frequency-locked loop*

  The frequency supplied to the DFT (fe), is estimated based on a double second-order generalized integrator frequency-locked loop (DSOGI-FLL) with gain normalisation as presented in [1]. For the single phase measurement, a SOGI-FLL is used for estimating the frequency.

  The following proportional gain parameters need to be defined:

- *SOGI proportional gain* (input parameter $Ksogi$)
- *FLL proportional gain* (input parameter $Gfll$)

In some cases, the frequency transients when locking the frequency can be very high, and therefore a rate limiter has been included in order to limit the estimated frequency. If the estimated frequency fe has high transients, a corresponding rate limit in Hz/s (parameter $rateLimit$) can be used in order to provide a "smoother" transient frequency value to the DFT calculation.

Please note that this option works very well also with unbalanced signals. The frequency tracking does not work properly for very strong distorted signals.

- *Fast fourier transformation*

The frequency supplied to the DFT is obtained by a Fast Fourier Transform (FFT) calculation (separately for each signal) at the clocked interval.

The following input parameters need to be defined:

- *Number of samples per period* (input parameter $samplesPerPeriod$)
- *Number of samples for FFT window* (input parameter $samplesPerWindow$)

The resulting sampling frequency in Hz, with which the calculation buffer is filled at defined intervals, is obtained using the nominal frequency $Fnom$ and the input parameter $samplesPerPeriod$ as:

$$samplingFreq = Fnom \cdot samplesPerPeriod \tag{73}$$

The resulting FFT window length in s of the buffer used in the calculation is obtained using the nominal frequency $Fnom$ as:

$$windowLength = \frac{samplesPerWindow}{samplesPerPeriod \cdot Fnom} \tag{74}$$

Using a moving FFT on the buffer gives us the frequency of the signal with which the DFT is executed.

Please note that this method has a limited bandwidth and may produce inaccurate results if the frequency deviates from the nominal frequency by more than $\pm 10\%$.

The outputs are updated at constant intervals defined by the parameter *Calculate outputs every* (input parameter $samplesForUpdate$) which defines after how many samples the DFT is calculated (and outputs are updated). The resulting update cycle (parameter $clockPeriod$) in s is:

$$clockPeriod = \frac{samplesForUpdate}{samplesPerPeriod \cdot Fnom} \tag{75}$$

When the option *Smooth outputs when updated* is used, the outputs are linearly interpolated (smoothed).

For the PQ measurement device, separately the RMS phasor values of the voltage and the current signals are measured and then the resulting active and reactive power quantities are calculated.

The available outputs for the DFT based measurement are presented in Table 4.2.

Table 4.2: Digital outputs, RMS phasor and symmetrical components (phase quantities) for EMT-Simulation

Name	Unit	Description
p_A	p.u.	RMS active power A
p_B	p.u.	RMS active power B
p_C	p.u.	RMS active power C
q_A	p.u.	RMS reactive power A
q_B	p.u.	RMS reactive power B
q_C	p.u.	RMS reactive power C
p	p.u.	Positive sequence active power
q	p.u.	Positive sequence reactive power
$p2$	p.u.	Negative sequence active power
$q2$	p.u.	Negative sequence reactive power
$p0$	p.u.	Zero sequence active power
$q0$	p.u.	Zero sequence reactive power

The available calculation parameters for the DFT based measurement are presented in Table 4.3.

Table 4.3: Calculation quantities, RMS phasor and symmetrical components (phase quantities) for EMT-Simulation

Name	Unit	Description
ur:A	p.u.	RMS voltage A, real part
ur:B	p.u.	RMS voltage B, real part
ur:C	p.u.	RMS voltage C, real part
ui:A	p.u.	RMS voltage A, imaginary part
ui:B	p.u.	RMS voltage B, imaginary part
ui:C	p.u.	RMS voltage C, imaginary part
ir:A	p.u.	RMS current A, real part
ir:B	p.u.	RMS current B, real part
ir:C	p.u.	RMS current C, real part
ii:A	p.u.	RMS current A, imaginary part
ii:B	p.u.	RMS current B, imaginary part
ii:C	p.u.	RMS current C, imaginary part
F:A	Hz	Frequency of the voltage signal A
F:B	Hz	Frequency of the voltage signal B
F:C	Hz	Frequency of the voltage signal C
Fi:A	Hz	Frequency of the current signal A
Fi:B	Hz	Frequency of the current signal B
Fi:C	Hz	Frequency of the current signal C

4.2.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase DFT based measurement are presented in Table 4.5.

Table 4.4: Digital outputs, RMS phasor and symmetrical components (phase quantities) for EMT-Simulation, single-phase measurement (*Number of phases*$= 1$)

Name	Unit	Description
p	p.u.	RMS active power
q	p.u.	RMS reactive power

The available calculation parameters for a single-phase DFT based measurement are presented in Table 4.5.

Table 4.5: Calculation quantities, RMS phasor and symmetrical components (phase quantities) for EMT-Simulation, single-phase measurement (*Number of phases*$= 1$)

Name	Unit	Description
ur	p.u.	RMS voltage, real part
ui	p.u.	RMS voltage, imaginary part
ir	p.u.	RMS current, real part
ii	p.u.	RMS current, imaginary part
F	Hz	Frequency of the voltage signal
Fi	Hz	Frequency of the current signal

A References

[1] R. Teodorescu, M. Liserre, and P. Rodriguez. *Grid Converters for Photovoltaic and Wind Power Systems*. Wiley, 2011.

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